UNIVERSITY OF HYDERABAD
SCHOOL OF ENGINEERING SCIENCES & TECHNOLOGY
CENTRAL UNIVERSITY P.O, PROF. C. R. RAO ROAD, GACHIBOWLI,
HYDERABAD –500046, INDIA

29/10/2025

Ref. No.: UoH/SEST/KVRSE/ARDB/DIC

Global tender enquiry for the supply of One 3D digital image correlation system.

The School of Engineering Sciences & Technology (SEST), University of Hyderabad (UoH), invites tenders in a two-bid format from highly reputed and established firms and authorized dealers to supply one 3D digital image correlation system which can operate at high temperatures also in addition to room temperature. Final Terms and Conditions are as per UoH's latest purchase policy. The tenders must reach the below given address below on or before 5.00 PM on November 21, 2025.

THE FOLLOWING CONDITIONS MUST BE STRICTLY MET IF THE QUOTATION IS TO BE VALID

1. All quotations should be addressed to below mentioned details and envelopes should be superscribed with the following: "One 3D, digital image correlation system":

   Prof. Koteswararao V. Rajulapati,
   School of Engineering Sciences & Technology,
   West Campus, University of Hyderabad Central University P.O.,
   Gachibowli, Hyderabad 500046, INDIA.

2. Quotations must reach before 5 PM on 21.11.2025. Quotations received after the due date and time will not be considered. The University of Hyderabad will not, in any way, be responsible for any postal or other delays.

3. The bids shall be submitted in a TWO-BID format in separate sealed covers. Item code, enquiry number & due date must be compulsorily mentioned on both envelopes contained within another main envelope duly sealed, which shall also have item code, enquiry number + due date superscribed.

   a) TECHNO-COMMERCIAL BID

   Please give detailed specifications, International standards (BIS INTERNATIONAL), catalogues, list of users & technical details, technical compliance sheet, etc., along with the operating parameters, pre-installation requirements, payment terms, warranty/guarantee period, delivery period, etc., very explicitly without any ambiguity. The quotation must clearly specify the make and model of the equipment. All relevant technical literature/brochures, application notes, and specifications must also be provided, adequately explaining and confirming compliance of the features of the model of the equipment being quoted with those given in the UoH tender enquiry. Without this, the quotation may be considered invalid. It must be clearly indicated if any of the features do not match. Explanations on the non-matching specifications will be considered. The

University of Hyderabad reserves the right to determine which deviations are substantial based on current and future requirements. If features in the quoted equipment are better than those indicated in the enquiry, they must also be clearly explained.

b) PRICE BID

Giving full prices in foreign currency or in Indian rupees for Main equipment Essential accessories & spares. All optional items Annual Maintenance Contract (AMC) charges for three years after the expiry of the warranty period.

4. **Price & insurance*:*** The price quoted should be as follows –
In case of Indian suppliers : Insurance and freight prepaid by the vendor, Free delivery at the School of Engineering Sciences and Technology, University of Hyderabad.
In case of foreign suppliers/import: CIP, HYDERABAD AIRPORT BASIS

   Note: The offers of the firms that quoted on Ex-works basis and do not mention the handling, documentation, packing, forwarding, transportation & insurance charges, etc., separately will not be considered as a responsive bid, and the acceptance of such bid will vest entirely with the University of Hyderabad.

5. **Vendor qualification criteria for acceptance of technical bid.**

   a) Vendor should have supplied at least 5 DIC Systems to Govt Organizations/Research Institutes in the past 5 years. Copy of Installation Report / Certificate must be submitted.
   b) Vendor must explicitly mention and provide proof of the items claimed under Make in India Scheme.
   c) Vendor must visit University of Hyderabad with their DIC System and demonstrate and validate the DIC measurements with available experimental data OR compare DIC Data with Strain Gage Extensometer and prove the accuracy of the DIC System. Vendor must also demonstrate High Temperature DIC upto 600 ºC. All travel costs to be borne by the vendor.

ALL OPTIONS SHOULD BE QUOTED SEPARATELY.

6. **Validity of prices:**
The quoted prices must be valid for a minimum period of 6 months from the date of opening of the bid, failing which offers may be summarily rejected by the University of Hyderabad.

7. **Compliance statement:**
   a. All manufacturers/agents must submit the compliance statement and the technical bid.
   b. Please provide the below details if you have supplied similar or identical items in India, particularly to other research/academic institutions in the last three years.
      - **Name of the item with brief specifications, model etc.,**
      - **Name & contact address of the user.**
      - **Any other relevant information.**

8. **Installation & commissioning:**
Installation & commissioning will be the sole responsibility of the supplier / Indian agent. The supplier must do installation & commissioning with all infrastructural works. All vendors/agents must submit full details and requirements for installation & commissioning of the items as per the technical specifications they submitted.

9. **Taxes, duties, and levies:**
The suppliers shall be solely responsible for all taxes, duties, levies, including Octroi, local levies, sales tax, and excise duties incurred as per extant laws in India. No concessional sales tax form "C" or

"D" will be issued by this University. It should be clearly mentioned in the quote whether the prices include taxes & duties. If the price includes taxes and a variation of tax element is sought, then the quote should mention the base level of taxes built into the price.

10. Customs and Excise Duty Exemption:

The University of Hyderabad is under the University Grants Commission (UGC), Government of India, and thus is exempted from excise duty vide GOI notification No.10/97 central excise dated March 1, 1997, and is eligible for concessional customs duty as per GOI notification No.51/96 customs dated 23/07/1996 – amended from time to time.

11. Delivery Earnest Money Deposit (EMD) / BidSecurity:

The bidders should furnish bid security for 5% of the estimated cost of the equipment along with their technical bid in the form of Account Payee Demand Draft / Banker's Cheque from any of the Commercial Banks in an acceptable form, drawn in favour of Finance Officer, University of Hyderabad, payable at Hyderabad. If the bidder furnishes a Fixed Deposit Receipt / Bank Guarantee, the instrument shall be issued in favour of Finance Officer, University of Hyderabad, payable at Hyderabad and shall be valid for a period of 135 days from the date of opening technical bids. The MSE units registered with MSME and certificates obtained from NSIC under the Single Point Registration Scheme (SPRS) shall be exempted from payment of Earnest Money Deposit (EMD) on production of requisite proof in respect of valid certification from NSIC.

12. Delivery locotion, Delivery Period, and Penalty Clauses:

- **a)** The items shall be delivered to: School of Engineering Sciences and Technology University of Hyderabad, Gachibowli, Hyderabad - 500 046, India.
- **b)** In case of foreign suppliers, the firm delivery period (in weeks) after opening an irrevocable Letter of Credit (L.C.) must be given in the Techno-commercial Bid. No part shipment is permitted.
- **c)** In the case of indigenous suppliers, the firm delivery period (in weeks) after the issue of the purchase order must be specified in the technical bid. Delivery is to be made at the University of Hyderabad. Delivery must be on a freight-paid basis for dispatch by road. University of Hyderabad will issue the certificate/declaration instead of a road permit to affect the delivery.
- **d)** In the event of delay in delivery beyond stipulated period indicated in the Purchase Order, liquidated damages @ 1% per each week of delay subject to a maximum of 5% of the cost of the order will be levied and collected by the University, by way of deduction from the payments due to the vendor/supplier.
- **e)** Delay in installation & commissioning beyond the contractual deadline will result in forfeiture of the performance bank guarantee.

13. Warranty:

The supplier must warrant that the goods supplied under the contract are new, unused, and the most recent or current, and incorporate all recent improvements in design and materials as per specifications in this tender document. All vendors (manufacturers/agents) must submit offers with 36 months of comprehensive warranty after installation & successful commissioning of the equipment at the School of Engineering Sciences & Technology, University of Hyderabad, Hyderabad-500046, India, which would be considered for the tender evaluation purpose. The supplier should, however, quote separately for a comprehensive warranty for three successive years after completing a three-year comprehensive warranty. (This is to get an estimate of the cost involved). The warranty offered should be unconditional, and the seller should agree to replace or repair the equipment at his cost. The spare parts for the warranty repair have to be provided by the vendor.

14. Payment :

- **a)** For Indian suppliers and manufacturers - 100% payment subject to submission of 10 %

performance bank guarantee (PBG) as per clause four will be made direct to Indian suppliers by "Account Payee" cheque against invoice within 30 days of delivery, inspection, successful commissioning & acceptance of equipment/services by the University of Hyderabad.

b) For Foreign suppliers, 100 % by irrevocable foreign letter of credit, subject to the vendor submitting an unconditional PBG from a scheduled bank of 10 % of FOB value, and before opening irrevocable L.C. This PBG must be valid for 60 days after the warranty period.

c) Intermediary agencies, local representatives of foreign manufacturers/suppliers, are involved in importing foreign goods, and an authorization letter/certificate must be obtained to this extent. No agency commission shall be paid to such intermediary agencies/local representatives.

d) In case of L.C.s, bank charges outside India, including confirmation charges, if any, shall be to the supplier's account. The documents for payment shall be presented to the purchaser's bank by the supplier within 21 days of shipment of the equipment.

e) An agreement per standard University norms to be entered with the University before the opening of L.C.

15. Agreement:

An Agreement in the format prescribed by the University of Hyderabad is required to be entered into by the Suppliers or their authorized Representative with the University in respect of purchase of one 3D, digital image correlation system. within a period of 15 days fromthe date of issue of Purchase Order.

16. Risk Purchase Clause:

If the supplier fails to deliver the ordered one 3D, digital image correlation system within the stipulated delivery period specified in the Purchase Order, the University may resort to procure such items and in such a manner as deemed appropriate, goods or services similar to those undelivered, and the Supplier will be liable to reimburse the University for any excess costs for such similar goods or services.

17. Risk to reject:

The University reserves the right to reject any or all the bids or accept a part of the quotation without assigning any reasons therefor. The decision of the University in this regard is final and binding on all the suppliers.

18. Resolution of disputes:

Any disputes arising out of this contract shall be referred to the University, and if any of the parties here to is dissatisfied with the decision, the dispute shall be referred to the decision of the Arbitrator, who should be acceptable to both the parties, to be appointed by the Vice- Chancellor of the University. The decision of such Arbitrator shall be final and binding on both the parties

19. Jurisdication of courts:

Disputes, if any, arising during course of execution of the order are to be settled within the jurisdiction of Hyderabad / Ranga Reddy District Courts only

20. Delay in submission:

The University will not in any way be responsible for any postal / courier delay. Bids received beyond the stipulated date and time of submission are summarily rejected. Tenders incomplete in any respect will be summarily rejected.

21. Opening of tenders / bids:

a) The bids will be opened at the School of Engineering Sciences and Technology, University of Hyderabad, Hyderabad in the presence of bidders or their authorised representatives.

b) The authorised representative shall bring the authorisation letter and submit the same to the University before opening of bids. On failure to do so, the Representative loses the right to participate in the bid opening process.

22. This procurement is entirely governed by the University of Hyderabad Purchase Policy (UHPP). The University of Hyderabad reserves the right to reject any/all quotations or accept any offer or part thereof, without giving any reasons. Its decision in the matter will be final.

23. Any clarifications regarding this tender enquiry should be sought from the undersigned before submission of a formal quotation.

Prof. Koteswararao V. Rajulapati
School of Engineering Sciences & Technology University of Hyderabad
Hyderabad 500046
E-mail:kvrse@uohyd.ac.in.

Technical Specifications

A digital image correlation system

The proposed 3D Digital Image Correlation (DIC) System should be a state-of-the-art, non-contact optical measurement solution for full-field strain and displacement mapping, which can operate at room temperature and high temperatures. The system should be designed to operate in compliance with **ASTM standards** for strain measurement, ensuring accuracy, reproducibility, and traceability of experimental data across a broad range of materials, including metallic alloys, ceramics, polymers, and composites etc.

At room temperature as well as under elevated temperature conditions, the system will enable high-fidelity strain field measurements, supporting both quasi-static and dynamic mechanical testing. The optical configuration, calibration procedure, and software algorithms ensure compatibility with existing infrastructure while maintaining measurement precision across varied test environments.

Importantly, the system shall be fully compatible with the existing Instron 5982 Universal Testing Machine and Zwick Vibrophore 100, thereby enabling direct integration for monotonic, cyclic, and fatigue testing applications. The architecture should be modular and upgradable, permitting future enhancements.

This purchase will establish a versatile, scalable, and future-ready experimental platform for advanced materials characterization, supporting both ongoing research and future demands in structural and functional material development.

Complete package with the following features:-

1. Image Capture:- Manual, Time, Streaming, Images in block as per frame rate, variable time capture and
Analog Capture.
2. Indication in color code if the image is focused and the noise levels.

3. Indication in color code if the image is uniformly illuminated.
4. Capture of external analog signals with the images – synchronization with the UTM.
5. Parameters to be Measured.
   a) Displacements - X, Y & Z Directions.
   b) Strains: $\acute{e}_{xx}$, $\acute{e}_{yy}$, $\acute{e}_{xy}$, Principal, Von Mises and Tresca strains. Resolution of Strain Measurement – 25 µm or better.
   c) Velocity Computation.
   d) In Plane rotations.
   e) Full Field Stress Computations.
   f) Poisson's Ratio.
   g) User Defined Variables.

6) Strain Computation Algorithms:- Lagrange , Hencky, Euler-Almansi, Log Euler-Almansi, Engineering and Biot.
7) Time Filters to filter out noisy data. Option should be provided to create separate Time Filtered Variable. For example, the user must have separate color plots of $\acute{e}_{xx}$ and time filtered $\acute{e}_{xx}$.
8) Smooth of Data.
9) Indication of the appropriate subset size to be selected.
10) Computation of statistical data: mean, median, min., max., standard deviation… etc.
11) Export of plot and data to MATLAB.
12) Export of data to FEM Softwares.
13) Extrapolation of the data from the correlated displacement / strain data to the edge of the AOI.
14) Real time strain measutrement and option for the output signal to give feedback to the machine should be included
15) High temperature filters with spares and high temperature paints have to be supplied
16) A dedicated computer with state of the art configuration and latest version of the software must be included.
17) Vendor must to demonstrate strain measurements using the DIC system as well as strain gauge. These two measurements should be in agreement.

Software licenses:

1. Portable USB License (Lifetime Validity) for – Image Acquisition Software + 3D Analysis + 2D Analysis Software must be provided.

Installation and training at no extra cost

The system should be installed and commissioned by the supplier at University of Hyderabad and give hands-on training and trial tests on the use of the system.